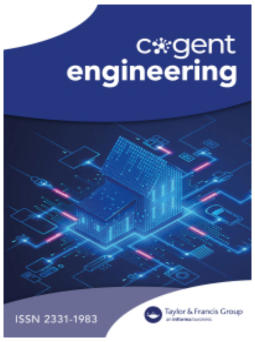

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Techno-economic viability and environmental sustainability of a grid-connected solar PV system for small and medium scale businesses in Cape Coast–Ghana using RETScreen expert

Kwame Anane-Fenin[a,b], Nana Yaa Serwaah Sarpong[a,b,c], Theophilus Adu Frimpong[d], Mukuna Patrick Mubiayi[e], Richmond Kuleape[f], Damenortey Richard Akwada[a], Wisdom Eyram Kwame Agbesi[a], Juliana Sally Renner[a], Charles Emmanuel Oppon[a,b], Lawrence Atepor[a] and Felix Menu[a]

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ABSTRACT
Ghana is home to some of West Africa's largest markets which have high energy consumption levels. This presents significant opportunities for solar energy investments. The study assesses the technical, economic and environmental viability of implementing three solar photovoltaic systems for three categories of small and medium-sized enterprises based on their energy demand. The results showed a promising energy yield of 1.42 kWh/yr per watt and a capacity factor of 16.3%. However, the economic analysis presented significant challenges such as a payback period ranging from 17.8 to 20.3 years, negative Net Present Value and levelized cost of energy higher than the base case grid electricity price. The environmental analysis demonstrated substantial potential for greenhouse gas emission reduction, with all systems achieving a 93% decrease compared to the local grid. A policy impact analysis was conducted to highlight industry-approved and research-based policies that meet global best practices.

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Photovoltaic systems; RET screen; renewable energy adoption; solar sizing; small and medium-sized enterprises (SMEs)

SUBJECTS
Hermodynamics; Heat Transfer; Design; Electrical & Electronic Engineering; Mechanical Engineering Design; Engineering Economics; Industrial Engineering & Manufacturing; Reliability & Risk Analysis; Power & Energy; Mechanical Engineering

1. Introduction

The global goal is to replace conventional energy sources (CES) with renewable energy sources (RES) to meet the growing global energy demand. The transition to 100% renewable energy (RE) remains the ultimate global agenda (Campagna et al., 2022; Iweh et al., 2023; Norouzi & Fani, 2021). At the same time, it aims to mitigate climate change, address the negative effects of greenhouse gas (GHG) emissions, protect the environment and ensure energy security. Renewable energy resources such as biomass, solar, wind and hydrogen fuels are being explored as alternatives (Spiru, 2023).

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International Renewable Energy Agency (IRENA) reports that the global capacity for generating electricity through renewable energy technologies (RET) was 2,356,346 MW in 2018. This indicates a consistent yearly rise of 8–9% during the preceding five years (Ruiz-Sánchez et al., 2022). According to estimates, 18.2% of the world's final energy consumption came from renewable energy in 2016, with 10.4% from modern renewable sources (REN21 Renewables Now, 2021). The total supply of energy from renewables has also been increasing, reaching 612 TWh in 2008, which represents a growth of 94% since 1973 (Elliott, 2019). In 2005, only 43 countries adopted any kind of RES targets, but by 2015, that number had risen to over 160 (Sawin et al., 2016). In Africa, there is a growing recognition of the need for renewable energy transition, as it offers cost-competitive and climate-compatible solutions. The potential for renewable energy is vast, with projections indicating that the region's renewable energy capacity will surge from 48.5 GW in 2019 to an estimated 169.4 GW by 2040 (Anane-Fenin, 2023). Despite the encouraging path toward renewable energy adoption, Ghana, like many other sub-Saharan African nations, grapples with energy crisis. These challenges are exacerbated by the fact that renewable energy consumption in most African countries remains minimal, with the continent heavily dependent on non-renewable energy sources (Mukoro et al., 2021).

According to the Energy Commission of Ghana, the country possesses an abundant solar energy resource that spans its entire territory. Daily solar irradiation levels range between 4 kWh/m^2 and 6 kWh/m^2, with annual sunshine durations varying from 1800 to 3000 h. This highlights a significant potential for grid-connected solar energy systems (Asumadu-Sarkodie and Asantewaa Owusu, 2016) (Sustainable Environment and Energy Systems, Middle East Technical University – Northern Cyprus Campus, Kalkanli, Guzelyurt, TRNC 99738/Mersin 10, Turkey, 2016). The distribution of solar intensities across Ghana is categorized as follows: the Savannah Zone, encompassing the Upper East, Upper West, Northern regions and northern parts of the Brong-Ahafo and Volta regions, experiences solar radiation levels between 4.0 and 6.5 kWh/m^2/day. The Middle Forest Zone, which includes the Ashanti, Eastern, Western and parts of Central, Brong-Ahafo and Volta regions, sees solar intensities ranging from 3.1 to 5.8 kWh/m^2/day. The Coastal Belt, covering Greater Accra, as well as coastal areas of Central and Volta regions, receives solar intensities ranging from 4.0 to 6.0 kWh/m^2/day (Asumadu-Sarkodie and Asantewaa Owusu, 2016) (Sustainable Environment and Energy Systems, Middle East Technical University – Northern Cyprus Campus, Kalkanli, Guzelyurt, TRNC 99738/Mersin 10, Turkey, 2016). This distribution emphasizes Ghana's potential for leveraging solar energy to diversify and expand its renewable energy portfolio. Figure 1(a,b) shows the direct normal solar irradiation and the Photovoltaic power potential of Ghana, respectively.

Ghana had a set target of 10% renewable energy (excluding large hydro) in the electricity generation mix by 2020, but this target was not met due to limited implementation of policies and a lack of market-driven support schemes (Onuoha et al., 2023). To address this, the government has extended the target to 2030 and developed a Renewable Energy Master Plan to guide investment and achieve the target (Asumadu-Sarkodie and Asantewaa Owusu, 2016) (Sustainable Environment and Energy Systems, Middle East Technical University – Northern Cyprus Campus, Kalkanli, Guzelyurt, TRNC 99738/Mersin 10, Turkey, 2016). Therefore, Ghana has the aim of increasing the renewable energy component of the national energy generation mix from 42.5 MW in 2015 to 1363.63 MW (with grid-connected systems totaling 1094.63 MW) by 2030 (Global Solar Atlas, 2024). The Renewable Energy Act, enacted in 2011, provides a regulatory framework and monetary incentives to promote renewable energy investment (Amo-Aidoo et al., 2022). This presents a good opportunity to increase its attractiveness to small-scale, residential and commercial consumers (Aboagye et al., 2021) either as Grid-tied systems connected or stand-alone systems isolated system with no interaction with a utility grid forming the two broad categories of photovoltaic (PV) systems. Small and medium-scale enterprises (SMEs) in Ghana can achieve significant cost savings by transitioning to renewable energy sources, but careful consideration of capacity factors, tariffs, system reliability and investment costs is necessary. However, the study by Opoku et al. highlighted several challenges faced by solar energy enterprises in Ghana, including limited resources for production and supply, high interest rates, inadequate incentives and a lack of technical know-how (Opoku et al., 2022; ITA, 2020). Therefore, while transitioning to renewable energy can lead to cost savings for SMEs, careful consideration of capacity factors, tariffs, system reliability and investment costs is necessary to overcome these challenges and ensure successful implementation (Appiah, 2022). Solar sizing for small businesses is a significant and extensively studied topic in renewable energy literature.

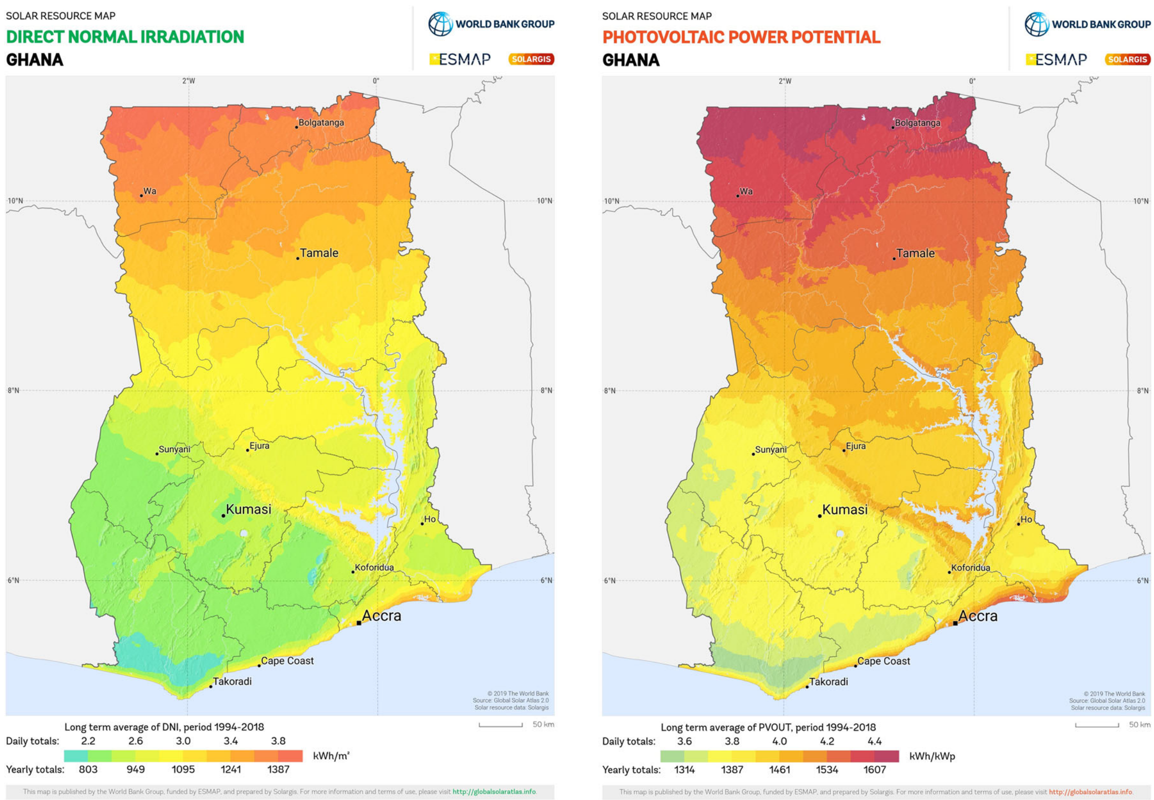

Figure 1. (a) Direct normal solar irradiation of Ghana (Global Solar Atlas, 2024). (b) Photovoltaic power potential of Ghana (Global Solar Atlas, 2024).

The involvement of small-scale enterprises in the renewable energy industry, particularly in small-scale embedded generation, hinges on the crucial aspect of solar sizing, as emphasized in a study by Mkhwebane and Ntuli (2019). Akram et al. also mention that optimal solar sizing plays a pivotal role in ensuring an economic and reliable energy supply for small businesses (Akram et al., 2017). They proposed an iterative search algorithm method to enhance efficiency and cost-effectiveness in solar system sizing. Moreover, Meunier et al. stressed the importance of considering various factors, such as load profiles, autonomy, self-consumption and investment capacity when sizing photovoltaic systems for small businesses (Meunier, 2015; Takase & Kipkoech, 2023). This comprehensive approach not only fulfills energy demands but also contributes to cost reduction and minimizes power supply loss, as highlighted by Babalola et al. (2014; Mkhwebane & Ntuli, 2019). Furthermore, Roberts discusses the significance of efficient solar sizing in achieving cost-effective energy solutions for small-scale businesses, particularly within the Solar Africa framework (Roberts, 2018; Akram et al., 2017). Additionally, Wedashwara et al. propose an innovative IoT-based Smart Small Scale Solar Energy Planning approach, facilitating energy efficiency determination crucial for businesses' sustainable energy planning efforts (Wedashwara et al., 2020; Meunier et al., 2015). Lastly, Ko introduces optimal sizing methodologies for solar water heating systems using genetic algorithms, showcasing their applicability for small businesses through an office building scenario (Ko, 2015; Babalola et al., 2014). In light of the insights gained from previous research and the multifaceted considerations in solar system sizing, it becomes evident that a holistic understanding of the complexities involved is crucial. This understanding paves the way for a more informed approach to designing and optimizing renewable energy systems for small businesses.

Sizing a system of this nature requires a careful understanding of several crucial factors. These factors play a vital role in determining the optimal configuration and capacity of the renewable energy system, ensuring it aligns with the specific needs of the business. Consideration of capacity factors, tariffs, system reliability and investment costs is crucial to designing an effective renewable energy system that optimally meets energy requirements while also being economically viable and environmentally responsible (Muazu et al., 2023; Roberts, 2018). Developing business-oriented systems requires knowledge of solar technology, availability of resources, optimal operation, economics, sizing, regulations, etc (Sandeep and Vakula, 2016; Wedashwara et al., 2020). A wide variety of simpler tools (INSEL, TRNSYS, PVSOL, SOLARPRO, HOMER, RETScreen, SOLinvest, Energy Periscope) exist for such purposes and mostly, system designers, installers, scientists and engineers use them for sizing, simulation, economic assessment and optimization of PV systems (Irwan et al. 2015; Ko, 2015). Of all these models, RETScreen is the most flexible in terms of the diversity of systems it can simulate and gives better results than other software available (Owolabi et al., 2019; Muazu et al., 2023; Sandeep & Vakula, 2016).

RETScreen is a powerful software designed for assessing and managing renewable energy projects (RETScreen International, 2024; Khandelwal & Shrivastava, 2017). It was developed by Natural Resources Canada's Canmet Energy Diversification Research Laboratory (CEDRL) in collaboration with partners such as the Renewable Energy and Energy Efficiency Partnership (REEEP), the National Aeronautics and Space Administration (NASA), the United Nations Environment Programme (UNEP) and the Global Environment Facility (GEF). RETScreen 4, an Excel-based software package, facilitates analysis of clean energy projects and allows users to develop energy-efficient project models and generate a five-step analysis report including energy usage, cost, emissions, financial benefits and risk associated with the project (Ganoe et al., 2014; Odoi-Yorke & Woenagnon, 2021). It covers a wide range of renewable energy sources like solar, wind and biomass and can analyze projects in grid-connected or stand-alone modes (Capareda, 2019; Afonaa-Mensah et al., 2024). The software calculates energy generation accurately by considering various factors (Mourouzi et al., 2018; Akuffo Paintsil et al., 2022). One of its standout features is optimizing the sizes of different components in renewable energy systems, making them more efficient and cost-effective (Jenkins and Ekanayake, 2017; Agyekum et al., 2019). It also supports comparing different project scenarios and ensures result accuracy through validation and verification. RETScreen can assess project finances, including costs and returns and evaluate their environmental impact (United States Environmental Protection Agency, 2020; Asamoah et al., 2023). The software is user-friendly and can be used by experts and non-experts in renewable energy.

However, RETScreen has some limitations. One significant drawback is its reliance on input data quality; inaccurate or outdated data can lead to unreliable results, particularly in regions with variable solar irradiance and climatic conditions (Nshimiyimana, 2018). Additionally, while RETScreen provides a comprehensive analysis of energy generation, financial viability and environmental impacts, it often lacks detailed sensitivity and risk assessments, which are crucial for understanding the uncertainties associated with renewable energy investments. Lastly, RETScreen may not adequately account for local regulatory frameworks and market dynamics, which can significantly influence project feasibility and success.

In this context, this study focuses on the practical application of these principles. By leveraging the capabilities of RETScreen, we delve into the economic assessment of grid-connected solar systems for small businesses in Cape Coast, Ghana. The research involves a thorough analysis of the economic feasibility of transitioning to renewable energy, incorporating cost-benefit analysis and payback period calculations specific to SMEs in Cape Coast. By considering the diverse energy needs and operational characteristics of these businesses, this study provides an understanding of the adoption of solar energy in small businesses. The methodological framework encompasses energy demand assessment, system sizing and economic assessment. This research transcends academia, offering practical guidance to investors and business owners considering renewable energy adoption in Cape Coast, Ghana. Utilizing RETScreen, we meticulously assess the economic feasibility of grid-connected solar systems tailored to small businesses' energy needs. In a world increasingly focused on sustainability, this study provides invaluable insights and a data-driven decision-making framework for those seeking sustainable opportunities. Additionally, our work extends to policy recommendations for the benefit of both businesses and the environment.

1.1. Literature on solar PV technology using RETScreen

The application of software tools for conducting solar PV analysis is frequently adopted for real-time and predictive studies. The most utilized tools include HOMER, HYBRID, PVSyst and RETScreen. RETScreen has been used to conduct several assessments of the financial viability of On and Off-grid PV systems of varied sizes and scale.

With the goal of transitioning from grid to 100% solar PV dependency, Rashwan et al. (2017; Irwan et al., 2015) using RETScreen evaluated the economic and environmental viability of residential accommodation in Saudi Arabia. A comparative analysis of the international PV model to electricity tariffs for government (4.0 cents/kWh) and the commercial sector (8.0 cents/kWh) was conducted. The study did not include sensitivity and risk assessment.

Another study carried out in Saudi Arabia using RETScreen aimed at assessing the feasibility of deploying a 10 MW grid-connected solar PV plant (Rehman et al., 2017). The comprehensive research utilized dry bulb temperature, global solar radiation, relative humidity and sunshine duration as input data for their model. The output parameters were Energy generated, greenhouse gas (GHG) emissions reduction potential, Net Present Value (NPV) and payback period. The simulation results were used in adopting a single solar PV facility.

Mehmood et al. (2014; Wahid et al., 2019) conducted a comparative analysis study on load correlation, solar irradiance and solar fraction RETScreen software for modeling a solar PV system for residential power load demand in Pakistan. An assessment of the modeled system incorporating technical, economic and environmental parameters such as GHG emission reduction potential, Internal Rate of Return (IRR), NPV and payback period. The results supported the feasibility of implementing a 5 kW off-grid solar PV system under ambient conditions within the project location. A predicted GHG emission reduction of 0.6–0.7 tCO_2 is expected. The objective of this study was focused on addressing a standalone system with a predefined load, therefore lacking the challenges associated with grid interconnectivity. Additionally, sensitivity and risk analysis were not included in the study.

An economic viability study on meeting power demand via renewable energy (RE) system design focusing on battery sizing and number of solar arrays was carried out using RETScreen software (Thevenard et al., 2000; RETScreen International, 2024). Based on climate, load and seasons, the optimal technology was determined through continued variation of system parameters. The authors did not factor risk and sensitivity analysis into the study.

Using RETScreen, Mirzahosseini & Taheri (2012; Ganoe et al., 2014) developed a model for assessing the technical, financial and environmental feasibility of solar PV plants in Iran. The researchers recommended a new tariff system using three different scenarios. However, GHG emission reduction credit scheme, risk and sensitivity analysis were not conducted.

A study conducted for a rural community in India used RETScreen to model a solar PV plant to evaluate its financial and environmental viability (Khandelwal and Shrivastava, 2017)(Capareda, 2019). The standalone plant supplied a load of 587.7 kW and, consequently, did not include the impact of grid interconnection on economic and environmental parameters.

From literature, several studies have been carried out in Ghana using other software such as HOMER (Odoi-Yorke and Woenagnon, 2021; Afonaa-Mensah et al., 2024; Akuffo Paintsil et al., 2022; Jenkins & Ekanayake, 2017; Mourouzi et al., 2018; United States Environmental Protection Agency, 2020). A number of studies using RETScreen software have been conducted in Ghana, however, these studies have mostly focused on other renewable energy sources such as wind, hydro and hybrid systems (Agyekum et al., 2019; Asamoah et al., 2023; Alhassan et al., 2023; Kofi, 2022; Agyemang-Boakye et al., 2024; Odoi-Yorke et al., 2023; Mehmood et al., 2014; Mirzahosseini & Taheri, 2012; Nshimiyimana, 2018; Rashwan et al., 2017; Rehman et al., 2017; Thevenard et al., 2000). Only a few Studies have been conducted on grid-connected solar PV systems using RETScreen in Ghana.

Agyekum et al. (2019; Nshimiyimana, 2018) used the RETScreen Software to evaluate the financial viability and feasibility of integrating a 2.5 MW solar power plant and wind energy into the energy mix in Ghana's southern, middle and northern sections. The northern belt recorded the highest solar irradiation levels of 6.07 kWh/m^2/day. It was recommended as the best location for siting such a project. A greenhouse gas (GHG) emission reduction of 93% will be the resultant impact of developing the plants. Although risk analysis was conducted, the study did not however include sensitivity analysis. The study was generally location-based and not geared toward a particular sector of the economy. Additionally, policy recommendations that could safeguard investment and mitigate risk were absent.

A study on the deployment of a floating solar photovoltaic (FPV) system comprising 500,000 solar panels projected to generate 520,233 MWh per annum was carried out on the Akosombo dam reservoir in Ghana (Alhassan et al. 2023; Rehman et al., 2017). The economic and environmental feasibility assessment using RETScreen software was conducted on the 420 MWp FPV plant with a capacity factor of 14,1%. The economic analysis showed a lower Levelized Cost of Electricity (LCOE) of $0.10/kWh, a payback period of 12 years and an annual GHC emission reduction of 308,904.5 tCO_2/MWh. The study included risk analysis; however, sensitivity analysis was not conducted. A comprehensive policy impact analysis was not included.

The two most recent studies that comprehensively assessed the financial and environmental viability of Solar PV systems were carried out in Nigeria and Cameroon. The study in Nigeria evaluated the feasibility of implementing a 6 [45] MW grid-connected solar system using the RETScreen tool (Owolabi et al., 2019; Muazu et al., 2023). The study critically assessed the technical, economic and environmental viability of deploying the solar PV plant in six locations within Nigeria. Risk and sensitivity analysis were considered in the study, however, policy impact analysis was notably excluded. The Cameroonian research is the most recent, which assessed the economic feasibility of implementing a 211.75 MW solar PV power plant utilizing RETScreen Expert software (Iweh et al., 2023). The simulation results showed an IRR-assets of 7.4%, which was lower than the debt interest and a cost-benefit ratio of 4.5, implying profitability. The study considered GHG emission reduction (61,004.5 tCO_2), risk and sensitivity analysis. The study did not include a policy impact analysis.

The current literature vaguely recommends some governmental policies that often lack an in-depth assessment of their impact. There is, therefore, a notable gap in recommending policies that meet best practices, industry standards and are research-based. Incorporating policy impact analysis is significant as it provides both the government and solar PV development investors with policy indicators that can influence the viability and profitability of such projects in Ghana.

1.2. Solar photovoltaic system

A solar photovoltaic (PV) system, also known as a solar power system, harnesses the energy of sunlight using PV modules to generate electricity. This electricity can be utilized immediately, fed back into the grid, or stored in batteries for later use. Solar PV systems are a dependable and eco-friendly power source, making them suitable for various applications within the residential, industrial and agricultural sectors (Chandel et al., 2014; Alhassan et al., 2023). The primary components of a solar photovoltaic (PV) system include solar panels, solar charge controllers, battery banks, inverters, auxiliary energy sources and loads. The selection of these components is based on the specific project requirements, site location and intended applications (Akinyele and Rayudu, 2013; Rashwan et al., 2017; Agyemang-Boakye et al., 2024; Kofi et al., 2022).

1.2.1. Solar panel

The solar panel is the essential component in photovoltaic (PV) systems responsible for converting solar energy from the sun into direct current (DC) electricity. These panels typically have a lifespan ranging from 20 to 30 years (Yekinni et al., 2023; Odoi-Yorke et al., 2023). Solar panels are utilized in various applications, including aerospace industries, electric vehicles, communication equipment, irrigation, aquaculture, water pumping and water heating. Due to rapid technological advancements and growing demand, the cost of PV panels has significantly decreased (Artaş et al., 2023; Chandel et al., 2014). PV cell technologies are categorized into six groups: monocrystalline silicon, polycrystalline silicon, amorphous silicon, cadmium telluride (CdTe), copper indium gallium selenide (CIGS) and polymer & organic PV.

1.2.2. Solar charge controller

The solar charge controller serves to manage the current flow from photovoltaic panels to batteries, thereby extending battery lifespan by preventing overcharging and over-discharging during operation. Its primary function involves determining whether solar panel-generated power is required for immediate load usage in low-voltage equipment and lighting or if it should divert excess power to charge deep-cycle backup batteries for future utilization (Rehman et al., 2017; Akinyele & Rayudu, 2013).

1.2.3. Solar battery

The solar battery functions as an energy reservoir to supply electrical appliances during periods of demand. Typically serving as a backup when solar intensity is insufficient to meet load requirements, it plays a crucial role in off-grid photovoltaic (PV) systems, which operate independently. In such systems, a battery ensures continuous power availability, sometimes supplemented by a generator. Conversely, for on-grid PV systems, battery backup is generally unnecessary as energy supplied to the grid directly correlates with solar array production, though exceptions exist where battery backup proves advantageous. Lead-acid batteries, with a nominal voltage of 2VDC, are widely favored for off-grid PV setups due to their durable design, minimizing electrolyte leakage from the terminals.

1.2.4. Solar inverter

The solar inverter serves as the central component of a solar power system, crucial for ensuring its safety, efficiency and reliability. It functions as an electronic device that converts the direct current (DC) output from photovoltaic (PV) panels into clean alternating current (AC) (Agarwal, 2022; Rashwan et al., 2017). Inverters are engineered to manage high voltages and currents under various environmental conditions, including high temperatures, moisture and voltage fluctuations. Typically, they are available in two main types: On-grid and Off-grid models (Ingewar et al., 2020; Yekinni et al., 2023). The ingress protection (IP) rating of an inverter denotes its ability to resist intrusion by foreign elements such as dust and water. Both the IP rating and the efficiency of the inverter are critical factors influencing its operational lifespan.

1.3. Ghana's energy policies and small businesses

The current energy policies and regulations in Ghana serve as both a catalyst and a challenge for Small and Medium Enterprises (SMEs). Ghana's commitment to addressing climate change is evident in its alignment with international agreements, such as the Nationally Determined Contributions (NDCs) and outcomes from the Conference of Parties (CoP) 26 (Ministry of Energy, 2021; Artaş et al., 2023). This commitment is captured in the National Energy Transition Framework, a comprehensive guide reflecting the country's dedication to evolving into a climate-resilient, low-carbon energy nation. This commitment has translated into the establishment of pivotal entities, including the National Energy Transition Implementation Committee and the National Energy Transition Coordinating Office. These entities, in collaboration with essential ministries such as the Ministry of Energy, Ministry of Transport and Ministry of Environment, Science, Technology, and Innovation have meticulously developed the framework through multiple stages, including desktop analysis, modeling, validation by regulatory bodies, collaborating ministries and stakeholder consultations (Ministry of Energy, 2021; Artaş et al., 2023). Furthermore, the government's emphasis on exploiting renewable energy resources is manifest in the legal foundation provided by the Renewable Energy Act, 2011 (Act 832) (Government of Ghana, 2011; Rehman et al., 2017) and associated sub-codes and guidelines. Tangible projects in clean cookstoves, alternative fuels, solar street lights and traffic signals, solar community lighting and mini-grids, underscore the practical application of this commitment (Ackah, 2017; Agarwal, 2022). However, other literature also highlights the prevalent challenge of energy poverty in Ghana, highlighting the imperative for effective energy policies and coping mechanisms for Small and Medium Enterprises (SMEs). These challenges, particularly in electricity supply and affordability, have tangible impacts on the operations and sustainability of SMEs (Arthur et al., 2023; Ingewar et al., 2020).

2. Methodology

2.1. Overview of methodology

The methodology for the renewable energy project comprises distinct phases as described in Figure 2:

- Energy potential assessment: Utilizes climatic data study to understand foundational aspects of the study area's renewable resources.
- Load demand assessment and categorization: Involves extensive surveys to precisely identify and categorize energy needs among targeted entities.
- Solar sizing: Utilizes RETScreen software for accurate and optimized configurations of the solar systems.
- Project viability analysis: Rigorously examines the project's economic feasibility using financial metrics, including simple payback, negative net present value, annual life cycle savings and benefit-cost ratios.

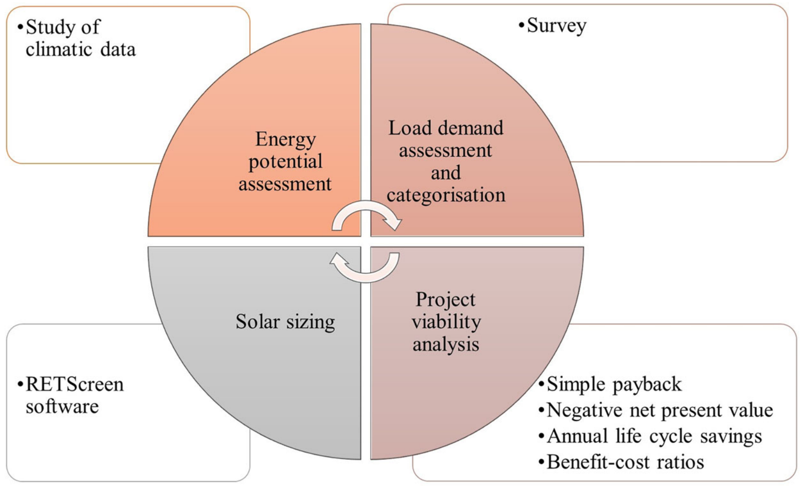

Figure 2. Methodological overview and approach.

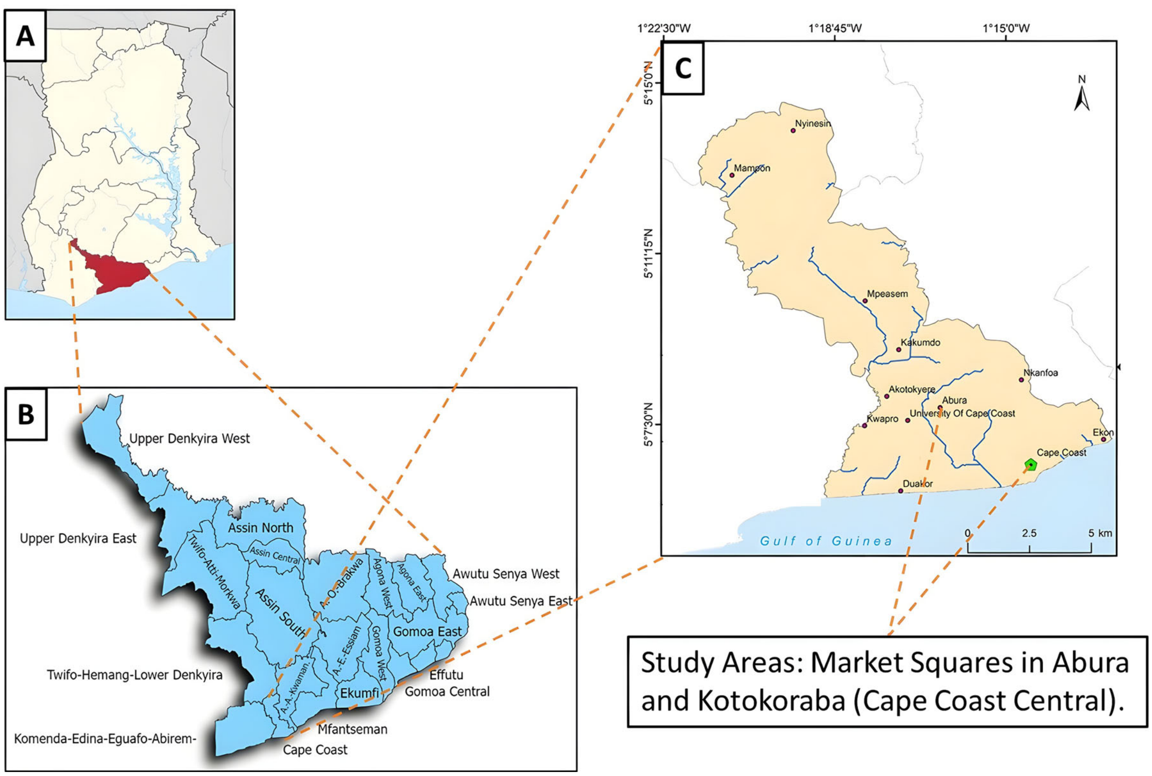

Figure 3. Map of the study area (A) map of Ghana, (B) map of the Central Region and (C) the Cape Coast Metropolis.

2.2. Insights into the study region: climatic data and energy assessment

2.2.1. Description of study area

Cape Coast is a historic city located in the Central Region of Ghana. It is a World Heritage Site and home to the famous Cape Coast Castle near the Gulf of Guinea. Cape Coast is located about 75 miles (120 km) southwest of Accra, the capital of Ghana, on a low point jutting into the Gulf of Guinea of the Atlantic Ocean. Two of its central market areas located at Abura and Kotokraba in the Cape Coast Metropolis were identified for this study and both lie at 5.13394° N, −1.27594° W and 5.1138° N, 1.2458° W, respectively. A thorough analysis of available solar energy resources potential in these selected sites in Cape Coast was conducted. Figure 3 shows the map of the selected study area areas.

2.2.2. Climatic data and energy potential of the study area

Cape Coast's climatic data reveals a generally favorable environment for solar energy generation. With consistently warm temperatures ranging from 23.9 °C in August to 27.5 °C in February, the region provides the ideal conditions for efficient solar panel performance. Additionally, the relatively low and stable wind speeds minimize the risk of damage to solar installations. While high relative humidity and significant rainfall during certain months may impact efficiency, the available solar radiation values of 4.04 to 5.37 kWh/m^2/day make Cape Coast well-suited for solar energy generation as seen in Figure 4.

Figure 4 singles out the monthly solar radiation values in Cape Coast is crucial for assessing the potential for solar energy generation in the region. It highlights significant monthly variations, with the lowest solar radiation in August (40.4 kWh/m^2/day) and the highest in March (53.7 kWh/m^2/day). These variations align with seasonal changes, with March being the end of the dry season and August in the rainy season. The annual average of 4.74 KWH/m^2/day underscores the region's solar energy potential, even during its lowest point as observed in Figure 5.

2.3. Assessment of energy demand patterns for SMEs in Cape Coast

2.3.1. The sampling technique and data collection

The sampling technique adopted for this study was the random technique where members have an equal opportunity of being selected from the population. A total of 500 small businesses were randomly selected, 250 from each central business center.

The sample size was obtained using the equation:

$$n = \frac{a^2bd}{e^2} \tag{1}$$

Where; n = sample size, a = level of confidence according to the standard normal distribution (95%), b = population having the characteristics sought, $d = 1 - b$, e = degree of precision.

After data processing and cleaning, 472 out of the initial 500 small businesses remained for the study. Thus, surveys and interviews were conducted on a representative sample of 475 SMEs to collect data

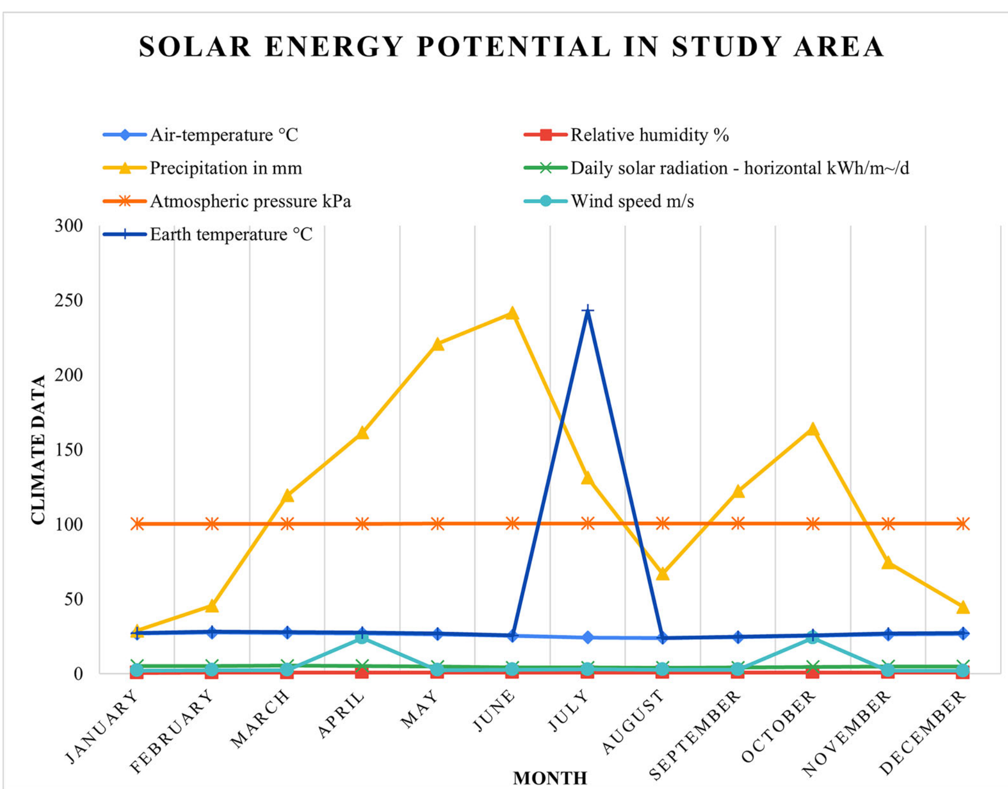

Figure 4. Climate data for the area under study.

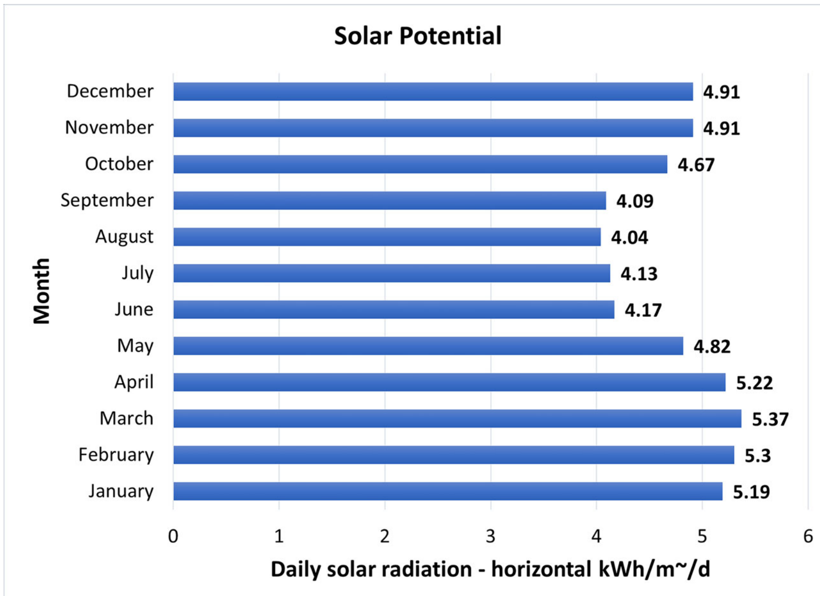

Figure 5. Monthly averaged daily solar radiation of Cape Coast.

from the two markets in Cape Coast to understand their energy demand patterns. The administered questionnaires were thoughtfully crafted to capture a range of pertinent data, including:

Biographic Data: Basic demographic information was collected to understand the profiles of the businesses and their owners.

- Load Demand: Detailed inquiries were made regarding the energy requirements and load demands of these businesses. This encompassed the identification of electrical appliances and equipment used in daily operations.
- Types of Appliances and Equipment: A comprehensive list of all electrical appliances, machinery and equipment used by the businesses. This should include their power ratings, operational hours and any special requirements.
- Operational Characteristics: Information on the operational characteristics of businesses, such as hours of operation and peak usage times, was meticulously documented.
- Energy Usage Patterns: Detailed data on how energy is used throughout the day, week and year. This includes information on peak usage times, off-peak periods and seasonal variations in energy demand.
- Energy Costs: Data on the current energy costs incurred by the businesses. This covers both grid electricity and any other energy sources they may be using, such as generators.
- Energy Bills: Copies of recent energy bills, if available, to provide insights into the historical energy consumption and costs.
- Installed energy meters at selected SMEs were monitored to obtain real-time energy consumption, peak demand and daily usage. The historical energy consumption data were obtained from the energy service providers, the electricity company of Ghana, ECG and the prices were also obtained from the Public Utilities Regulations Commission, PURC.

2.4. Solar sizing in RETScreen environment

Solar sizing is a crucial step in designing a photovoltaic (PV) system to harness solar energy effectively. RETScreen requires six types of data for simulation and optimization including meteorological data, load profile, equipment characteristics, search space, economic and technical data. These data are described in detail in the following subsections

- Meteorological data: To assess renewable energy potential, RETScreen needs local meteorological data. This data includes information on solar radiation, wind speed, temperature and other relevant weather conditions. Accurate meteorological data is crucial for realistic energy production estimations.
- Load profile: RETScreen requires data on the electrical load profile, which outlines how energy is consumed over specific time intervals. This data helps in understanding when peak loads occur and how energy is used throughout the day, week and year.
- Equipment characteristics: Detailed information about the characteristics of renewable energy equipment is vital. This includes specifications for solar panels, batteries, inverters and any other components used in the system. Knowing the efficiency, capacity and other technical details is essential for accurate simulations.
- Search space: RETScreen relies on user-defined parameters that dictate the range of potential system configurations. These parameters create a search space within which the software explores different options to find the optimal system design.
- Economic data: Economic data encompasses financial information related to the project, including costs, financing options, incentives and operating expenses. This data helps in evaluating the economic viability of the renewable energy project.
- Technical data: Technical data covers various aspects of the project, such as installation and maintenance costs, expected lifespan of equipment and any technical constraints that may impact the system's design and operation.

These parameters provide the foundation for RETScreen's built-in photovoltaic project model, which simulates system performance, estimates energy generation and assesses financial and environmental outcomes. Following the simulation, RETScreen generates detailed reports for analysis and interpretation. These reports typically include energy analysis (estimating annual or lifetime energy production), cost analysis (detailing investment, operation and maintenance costs), financial analysis (assessing project viability using metrics like NPV and IRR) and emission analysis (estimating GHG emission reductions compared to conventional electricity sources). Additionally, RETScreen offers valuable risk and sensitivity analysis tools. By allowing users to explore how variations in input parameters impact project outcomes, these tools empower informed decision-making regarding solar PV system adoption.

Solar sizing at Level 2 involves a more detailed analysis than a preliminary assessment but is still somewhat simplified compared to a comprehensive Level 3 analysis. Level 2 solar sizing focuses on achieving a reasonably accurate estimate of the solar system's size and energy production potential and was used in this study. The technical inputs provided by the RETScreen software, along with the product list and pricing data for various components, play a pivotal role in the comprehensive design and cost evaluation of the solar PV system. These inputs encompass critical elements such as the type and efficiency of the photovoltaic panels, power capacity, manufacturer and model specifications and the number of units, ensuring the selection of reliable and efficient components. Moreover, factors like temperature coefficients, solar collector area, bifacial cell adjustment and losses are meticulously considered to optimize system performance and sizing.

The data on inverter efficiency, capacity, initial and maintenance costs and electricity export rates shed light on the operational and economic aspects, facilitating a holistic approach to solar system design and cost assessment. This integrated data ensures that the final photovoltaic system configuration is not only technically sound but also economically viable, aligning with energy needs and budget constraints in the area of study. Table 1 shows components and specifications were chosen as seen in the RETScreen Energy Model environment.

2.5. Optimal sizing and configuration of solar PV systems

A comprehensively perform an optimum sizing a select the best Solar PV system configuration, energy audits were performed on selected SMEs. The energy audit indicated that the active operation time of the SMEs is within the day (between 8 am and 5 pm), the suitable solar PV system configuration to be considered is the grid-tied Solar PV system. In the grid-tied system configuration, surplus solar energy

Table 1. Selected components and specifications.

Photovoltaic	
Type	Poly-Si
Manufacturer	Jinko Solar
Capacity per unit	300 W
Model	Poly-si - JKM300P-72
Efficiency %	15.46%
Nominal operating cell temperature °C	45 °C
Miscellaneous losses %	3%
Frame area m^2	1.94 m^2
Inverter	
Efficiency %	95%
Capacity %	60% of power capacity
Miscellaneous losses %	3%
Summary	
Capacity factor %	16.3%
Initial costs $kW	2100 $/kW.
O&M cost (savings) $kW-year	21 $/kW-year.

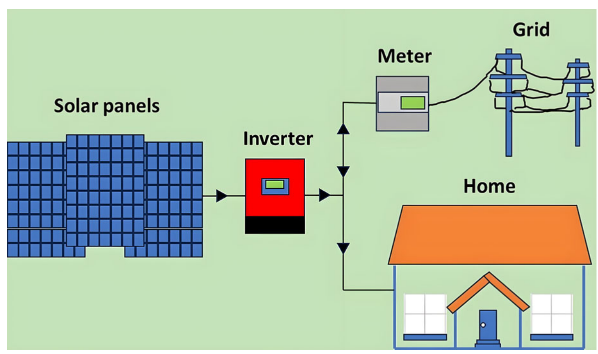

Figure 6. Grid-connected solar system without energy storage.

generated during periods of abundance is directed into the grid, offsetting the user's grid-based electricity consumption as shown in Figure 6. During periods of limited solar energy generation, the system relies exclusively on grid-supplied electricity to meet energy requirements. This configuration is noted for its cost-effectiveness and suitability in both residential and commercial settings and hence adopted in this analysis.

2.6. Load assessment and categorization approach

The surveyed businesses comprised different equipment such as fans, fridges, hair dryers, fans, computers, televisions, radios, etc., which contributed to the total demand. Since these equipment types are of different brands and efficiencies, it became necessary to assume an average equipment power rating to estimate the energy requirement of all the SMEs. Table 2 shows the data on the electrical components used in a single facility, their power ratings and the estimated energy requirements.

The size of the solar PV system (PV capacity) is determined by:

$$PV\ capacity = \frac{\mathrm{E/daily}}{S_h.Tloss.df.\eta inv} \tag{2}$$

Where: $E/_{daily}$ = electricity consumption per day, S_h = peak sun hours, T_{loss} is the temperature loss, df = derate factor of the system, η_{inv} = the inverter efficiency.

The peak sun hours for Cape Coast is 4.5 h and temperature loss (accounting for the impact of temperature on the system's performance since solar modules are less efficient at high temperatures) is 0.88. The inverter efficiency for the selected is 0.95. The derate factor (0.80) accounts for all losses in the system. For grid tie system, it is the product of the cable losses and the efficiency of the inverter.

$$fd = \eta inv(mppt) * \ \eta temp, mismatch * \eta cabbles \tag{3}$$

Table 2. Energy load assessment data for a surveyed business.

Item	Power range (W)	Total power (W)	Demand factor	Time of use
Television set	20–400	80	1.0	7
Fan	60–90	75	0.7	10
Light bulbs	20–180	50	1.0	10
Hair dryer	800–200	*1200*	0.3	5
Refrigerator	100–2000	150	0.8	24
Air conditioner	500–1500	*2500*	0.8	7
Desktop computer	200–800	300	0.9	7
Laptop computer	20–100	50	0.9	7
Security cameras and DVR/NVR systems	20–150	75	1.0	24
Audio systems:	50–500	200	0.8	7

Where the efficiency of the selected inverter at maximum power point is 95%, a temperature mismatch loss of 12% and the transmission losses for this design is kept at 4%

$$fd = 0.95 * 0.88 * 0.97 \quad = \quad 0.80$$

this value collaborates the sizing verification study by (Omar & Shaari, 2009).

Following the load assessment, the energy needs of small businesses were determined and categorized into three distinct groups based on their operational characteristics and energy demand. This step ensured that solutions would be tailored to the specific requirements of each category. These categories are tagged with tiers, namely Low Load Demand (Tier 1), Moderate Load Demand (Tier 2) and High Load Demand (Tier 3), which were established to account for the wide spectrum of business types present and their varying electrical needs. This tiered approach ensured that the diversity of businesses and their associated energy requirements were accurately represented.

2.7. Assumptions in the simulation

The study categorizes small and medium-sized enterprises (SMEs) based on their energy consumption levels, which informs the sizing of the solar PV systems. Also, daily solar irradiation levels in Ghana, ranging from 4 kWh/m^2 to 6 kWh/m^2, were assumed to optimize energy generation potential. A miscellaneous loss of 3% was assumed for all components, reflecting realistic operational conditions for solar systems in Ghana.

The study incorporates existing policies and regulatory frameworks aimed at promoting renewable energy adoption in Ghana, which influences economic viability. Also, the study assumes that the project would benefit from a concessional loan incentive (at a debt interest rate of 8%) backed by government support for renewable energy projects. The cost of the system, including initial capital costs, operational and maintenance costs, are all related to only grid-connected rooftop systems as investigated in the study.

2.8. Economic viability analysis of the proposed SMEs energy source transition

The costs and benefits associated with transitioning to a Solar PV system, considering initial setup costs, maintenance and potential savings, were evaluated using the RETScreen software tool. A survey was conducted on the financing options suitable for them and majority (about 80%) opted for debt equity financing in the proportion of 60% financed through debt and the remaining 40% comes from equity. The economic viability of grid-connected solar PV adoption with the current reliance on grid electricity and other backup sources was compared. Various economic indicators are frequently employed when assessing the viability of installing solar PV systems. These indicators help decision-makers evaluate the financial aspects of such projects. The payback period, Net Present Value (NPV) and Life-Cycle Cost (LCC) for the investment in grid-tied systems were determined by comparing the costs with the anticipated savings over time.

Payback Period (PP): The payback period for a solar PV system is a fundamental metric, computed as the initial investment cost divided by the first year's revenues generated from energy savings or production. A shorter payback period is often associated with lower perceived risks. This analysis focuses primarily on initial costs and current energy savings.

$$Payback\ Period\ (PP) = \frac{Initial\ Investment\ Cost}{Annual\ Savings\ or\ Revenue} \tag{4}$$

Net Present Value (NPV): Net benefit analysis, commonly expressed as NPV, quantifies the net difference between the benefits and costs of the solar PV system compared to an alternative, expressed in terms of present or annual monetary value. A system is considered cost-effective if its NPV is positive.

$$NPV = \sum_{t=0}^{n} \frac{Rt}{1+r} \tag{5}$$

Where: Rt = net cash inflow in year t, r = discount rate (interest rate), n = number of years, C = initial investment cost.

Life-Cycle Cost (LCC) Analysis: In LCC analysis, all relevant present and future costs associated with a PV system are summed up in present or annual value throughout the system's lifespan. These costs encompass energy, installation, operation and maintenance, repairs, less salvage value, inflation and discount rates for the investment's lifespan. Comparing the life-cycle costs of a rooftop solar PV system to alternatives helps determine its cost-effectiveness.

$$LCC = Initial\ Investment\ Cost + \sum_{t=1}^{n} \frac{Ct}{(1+r)t} \tag{6}$$

Where: Ct = cost or savings in year t, r = discount rate, n = number of years in the system's lifespan.

To effectively calculate these parameters, the following input parameters were used and specific numbers chosen from the year 2023 are detailed in Table 3.

Inflation Rate: The inflation rate reflects the expected general increase in prices over time. It is an important factor to consider when assessing the financial viability of a project, as it affects both costs and revenues.

Discount Rate: The discount rate is used to calculate the present value of future cash flows. It is a measure of the time value of money and is typically used to evaluate the net present value (NPV) of a project. A higher discount rate means that future cash flows are worth less in today's terms.

- Reinvestment Rate: The reinvestment rate is the rate at which any positive cash flows are reinvested. This rate can impact the overall financial performance of the project, particularly in cases where there are returns on investment that can be reinvested.
- Project Life: The project life is the duration for which the project is expected to operate. It's a critical parameter for assessing the long-term financial returns of the project.
- Debt Ratio: The debt ratio represents the proportion of project financing that comes from debt as opposed to equity. A higher debt ratio means a larger portion of the project is financed through borrowing, which can influence project risk and financial structure.

2.9. Environmental impact analysis

The Greenhouse Gas Emission Reduction Analysis is a crucial component within the RETScreen Clean Energy Project Analysis Software, designed to estimate the mitigation potential of greenhouse gas emissions in proposed projects. This analysis primarily focuses on assessing the environmental advantages of

Table 3. Financial parameters from 2023.

Parameter	Figure
Inflation rate	11%
Discount rate	21%
Reinvestment rate	32%
Project life	25 years
Debt ratio	60
Debt interest rate	8
Debt term	13

Sources: Bank of Ghana (2023), Enbar et al. (2016), Georgitsioti et al. (2019) and "Ghana: Electricity Prices" (2025).

transitioning from a baseline system powered by natural gas to a proposed system leveraging solar energy. The comprehensive dataset utilized in this analysis incorporates factors related to fuel types, electricity mix, emissions factors, GHG equivalences, energy generation efficiency and transmission and distribution losses. Emission factors, such as those for carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O), are sourced from reputable references like the Intergovernmental Panel on Climate Change (IPCC) 2014 guidelines, providing a foundation for quantifying direct emissions from each energy generation system.

The analysis proceeds to calculate emissions per megawatt-hour (MWh) of electricity generated by both the baseline and proposed systems, with a strong focus on electricity generation efficiency. The use of Global Warming Potential (GWP) values facilitates the conversion of GHG emissions into equivalent CO_2 emissions, enabling standardized comparisons across different greenhouse gases. The incorporation of Transmission and Distribution (T&D) losses, set at a fixed rate of 7.0% for both systems, considers energy loss during the transmission and distribution of electricity. Ultimately, the results of this analysis provide a critical assessment of each system's environmental performance, with a lower emissions profile indicating a more sustainable and environmentally friendly choice. These findings serve as a foundation for informed decision-making and support the reduction of GHG emissions, thereby promoting the adoption of cleaner, sustainable energy sources such as solar power.

2.10. Sensitivity analysis

In RETScreen, sensitivity analysis is conducted to assess the impact of variations in key input parameters on project outcomes. Initially, a baseline scenario is established using default or initial values for all inputs. Then, individual parameters such as capital costs, operating expenses, discount rates and energy production assumptions are systematically adjusted within plausible ranges. Sensitivity analysis helps identify which parameters have the most significant influence on project economics, aiding in risk assessment and decision-making. This process involves recalculating project metrics such as net present value (NPV), internal rate of return (IRR) and payback period for each variation. In this study sensitivity analysis were performed impact of parameters like debt interest rate, electricity export to the grid and discount rate on the NPV.

3. Results and discussion

3.1. Technical performance analysis

This section provides a concise analysis of the technical performance of the proposed photovoltaic (PV) systems. Table 4 presents the annual energy yield for all the Tiers considered for the three PV systems.

The simulated energy yield from the proposed systems gives an annual per-watt yield of 1.42 kWh/yr with a capacity factor of 16.3%, indicating a good system sizing with an expected better performance ratio when installed.

3.2. Cost estimate of proposed tier categories

The cost analysis data is presented for three solar power system capacities: 1.8 kW, 3.3 kW and 0.6 kW in Table 5. The costs are categorized into various phases, including feasibility study, development, engineering, power system, balance of system, O&M and periodic inverter replacement over 13 years. The initial costs range from $2125 for the 0.6 kW system to $*6110* for the 1.8 kW system, with the 3.3 kW system falling in between at $11,172.5.

Table 4. The annual energy yield for all the tears considered for the three PV systems.

Tier	Installed PV capacity (kWp)	Yearly energy produced (kWh/yr)
Tier 1	0.6	855.4
Tier 2	1.8	*2572.5*
Tier 3	3.3	*4702*

Table 5. Cost estimate for the project life of proposed tier categories.

	System sizes		
Parameters	Tier 1 (0.6 kW)	Tier 2 (1.8 kW)	Tier 3 (3.3 kW)
Feasibility study ($)	10	20	30
Development ($)	10	20	30
Engineering ($)	10	20	30
Power system ($)	1260	*3780*	6930
Balance of system and miscellaneous ($)	220	220	220
O&M - 25 years ($)	315	1050	1732.5
Periodic cost; inverter replacement - 13 years ($)	300	1000	2200
Total ($)	2125	6110	11,172.5

Table 6. Financial summary for different tiers of solar photovoltaic systems.

Parameters	Tier 1 (0.6 kW)	Tier 2 (1.8 kW)	Tier 3 (3.3 kW)
Total initial cost ($)	1480	4000	7150
Annual cost and debt payment ($)	125	341	612
Total annual savings and revenue ($)	86	257	470
Net yearly cash flow ($)	−39.4	−84	−142

Table 7. Financial viability analysis of solar power systems.

Parameter	Tier 1 - 0.6 kW	Tier 2 - 1.8 kW	Tier 3 - 3.3 kW
Simple payback yr	20.3	18.3	17.8
Equity payback yr	None	None	None
Net present value (NPV) $	−851	−2263	−4134
Benefit-cost (B-C) ratio	−0.44	−0.41	−0.45
Annual life cycle savings $/yr	−180	−479	−876
Energy production cost $/yr	0.476	0.331	0.298

3.3. RETScreen financial analysis

The data presents the financial analysis of three solar power systems with capacities of 0.6 kW, 1.8 kW and 3.3 kW in Table 6. Total initial costs for the systems range from $1480 to $7150, covering all setup expenses. Annual costs and debt payments vary from $125 to $612, while total annual savings and revenue generated by these systems range from $86 to $470. The net yearly cash flow, representing the difference between annual expenses and savings, is negative for all three systems, indicating that expenses currently outweigh income. These results suggest that these systems may not be financially viable in their current state.

3.4. Financial non-viability of proposed tiers

The provided financial viability data in Table 7 for three tiers of solar power systems (0.6 kW, 1.8 kW and 3.3 kW) reveals a challenging investment landscape for an investor. All three tiers exhibit negative Net Present Values (NPV), indicating that the initial investment costs outweigh the expected future savings. The annual life cycle savings are also negative, signifying that these systems are not generating cost savings but rather incurring ongoing costs for the investor. The simple payback period for each tier ranges from 17.8 to 20.3 years, which is quite lengthy and may be perceived as a high-risk investment with prolonged returns. Additionally, the equity payback years are marked as 'None', suggesting that equity is not recovered within the given time frame. In summary, this data implies that these solar power systems are not financially attractive and investors may need to reconsider the investment or explore alternative options to achieve a more favorable financial outcome.

3.5. System targets and cashflow

The cumulative cash flows are plotted versus time in the cash flows graph in Figure 7. These cash flows over the project life are calculated in the model and reported in the Yearly Cash Flow graph. A positive cumulative cash flow indicates the power plant is generating more cash than the amount spent. The initial years of the cumulative cash flow are negative, which indicates that the amount spent on the system

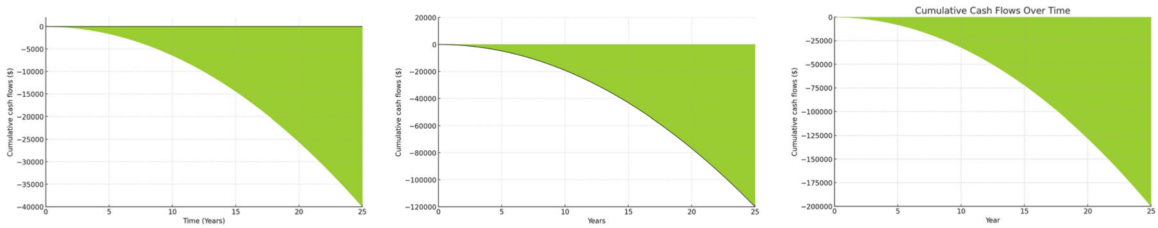

Figure 7. Cash flow diagram of various tiers.

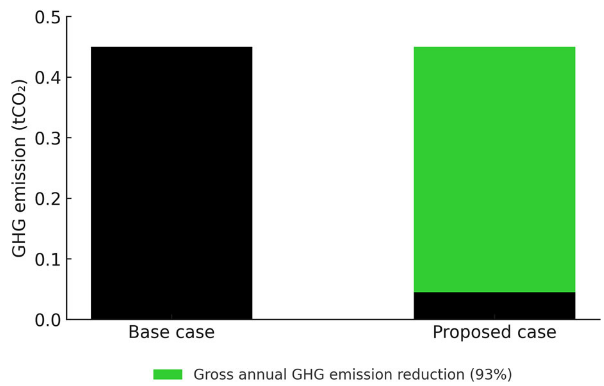

Figure 8. Graph of GHG emission of base and proposed case.

is more than the amount being generated by the system. At the beginning of the project, there is typically a significant cash outflow, representing the initial investment required to start the project. This is why you see the negative portion at the start of the graph. In the early years, operational expenses, such as maintenance, operating costs and debt payments, may outweigh the revenues or income generated by the project. This results in a negative cash flow, which is the shaded portion in the negative zone. However, the expansion of the negative zone in a cash flow diagram in all three scenarios, particularly after 14 years, signifies that the project is incurring escalating losses or cash outflows during this period. This observation raises concerns about the project's financial health and resilience. Implications include heightened financial risk, a prolonged payback period, a need for re-evaluation to address underlying issues, potential impacts from market or economic conditions and the risk of eroding investor confidence. To address these challenges, it's imperative to analyze the root causes, implement strategic adjustments and ensure sustainable financial performance to secure the project's long-term viability.

3.6. GHG emission analysis

In a GHG gas emission analysis in RETScreen, the provided data demonstrates the remarkable environmental benefits of adopting different tiers of solar power systems (see Figure 8). The baseline GHG emissions (measured in metric tons of carbon dioxide, tCO_2) for Tier 1 (0.6 kW), Tier 2 (1.8 kW) and Tier 3 (3.3 kW) are 0.4 tCO_2, 1.2 tCO_2 and 2.2 tCO_2 per year, respectively. After implementing these solar systems, emissions are drastically reduced, with Tier 1 achieving a complete offset and Tier 2 and Tier 3 reducing emissions to 0.1 tCO_2 and 0.2 tCO_2 per year. This translates to a substantial 93% reduction in GHG emissions for all tiers, underlining the positive environmental impact of using solar power for electricity generation. Furthermore, the data presents this reduction in terms of the equivalent number of cars and light trucks not being used, making the emissions reductions readily understandable and relatable.

3.7. Sensitivity and risk analysis of the proposed solar PV sys for Tier 2 business

The sensitivity and risk analyses of the proposed Tier 2 Solar PV project yield critical insights for Small and Medium Enterprise (SME) adoption. The sensitivity analysis reveals persistently negative Net Present

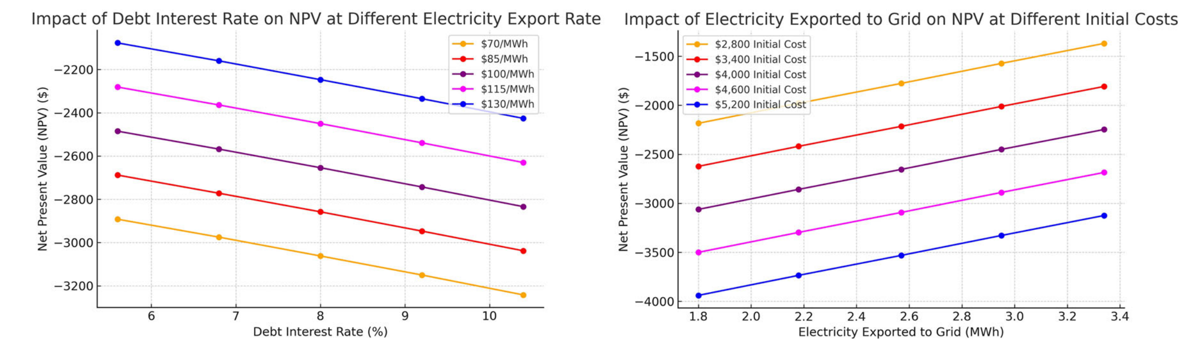

Figure 9. Sensitivity analysis with (a) debt interest rate and (b) electricity exported rate of the proposed solar PV system for Tier 2 business.

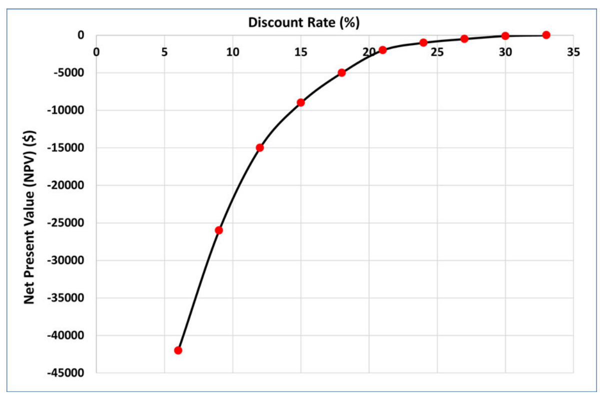

Figure 10. Sensitivity analysis with discount rate of the proposed solar PV system for Tier 2 business.

Values (NPV) across all scenarios, suggesting potential financial non-viability under current conditions, as demonstrated in Figures 9(a,b) and 10.

The analysis demonstrates an inverse relationship between interest rates and NPV, with higher rates diminishing project attractiveness. As the debt interest rate increases, the NPV consistently decreases across all electricity export rates. For instance, at a lower debt interest rate (5.6%), the NPV is higher for all electricity export rates (eg $70/MWh has the highest NPV of −$2892, while $130/MWh has −$2077). Also, at a higher debt interest rate (10.4%), the NPV decreases significantly (eg $70/MWh has −$3242, while $130/MWh has −$2426). This indicates that higher debt interest rates increase the cost of financing, making the project less profitable. Conversely, lower interest rates improve NPV, particularly when combined with higher electricity export rates. This highlights the importance of securing favorable financing terms to enhance the financial viability of solar PV projects.

Figure 9(b) shows the impact of electricity exported to the grid on the NPV of the proposed Solar PV system for Tier 2 businesses. The analysis reveals that as the electricity exported to the grid increases, the NPV improves across all initial cost scenarios. For instance, at 1.8 MWh exported, the NPV is lowest (eg −$2623 for $3400 initial cost and −$3939 for $5200 initial cost). Also, at 3.34 MWh exported, the NPV improves significantly (eg −$1808 for $3400 initial cost and −$3124 for $5200 initial cost). This suggests that exporting more electricity improves NPV by increasing revenue. Therefore, optimizing system efficiency and ensuring maximum energy export to the grid can significantly boost the project's financial returns. Additionally, higher initial costs reduce NPV across all levels of electricity exported, emphasizing the need to lower initial costs through subsidies, tax benefits, or optimized design.

This discount rate analysis highlights the importance of carefully managing financial and operational parameters to maximize the profitability of the proposed solar PV system. The curve shows that as the discount rate increases, the NPV initially decreases, remaining negative until it reaches approximately a discount rate of 27%. Beyond this point, the NPV turns positive, indicating that higher discount rates result in a greater return on investment. The critical point occurs around a discount rate of 28% (as the

Risk analysis

Perform analysis on:	Energy Production cost
Number of combinations:	500
Random seed:	No

Parameter	Unit	Value	Range (+/-)	Minimum	Maximum
Initial Costs	$	4,000	25%	3,000	5,000
O&M	$	38	25%	28	47
Electricity exported to grid	MWh	2.57	25%	1.92	3.21
Electricity export rate	$/MWh	100	25%	75	125
Debt ratio	%	60.00%	25%	45.00%	75.00%
Debt interest rate	%	8.00%	25%	6.00%	10.00%
Debt term	yr	13	25%	10	16

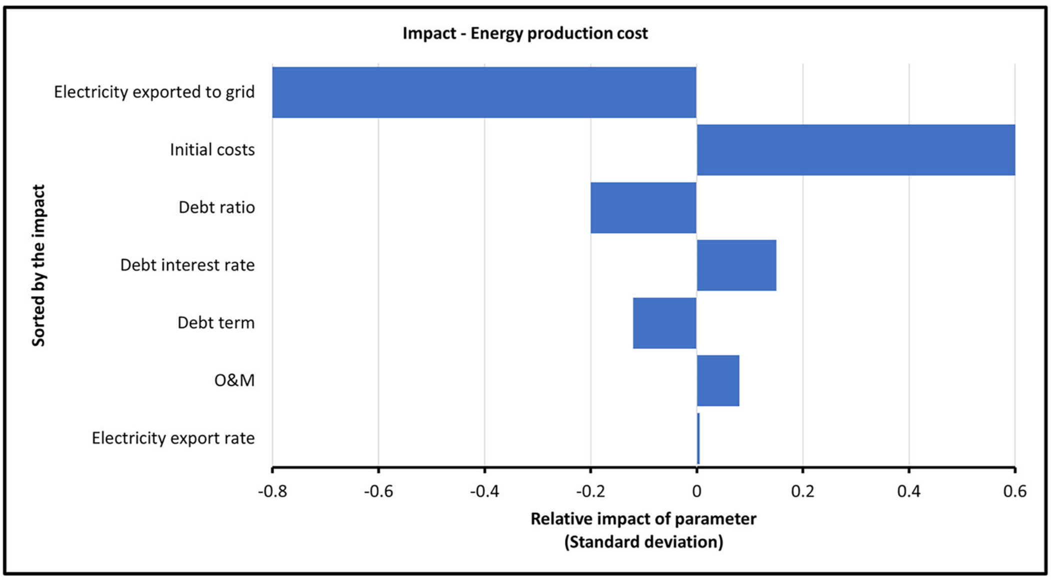

Figure 11. Risk analysis of the proposed solar PV system for Tier 2 business.

break-even point), where the project shifts from being unprofitable (negative NPV) to profitable (positive NPV). The steep decline in NPV at lower discount rates reflects higher risks associated with future cash flows. A project with high discount rates as less desirable due to perceived risks and uncertainties. However, if stakeholders expect a required return greater than 27%, they may reconsider investing in this project since it yields a negative NPV at lower rates.

The risk analysis (see Figure 11), focusing on energy production costs, identifies electricity exported to the grid as the most significant factor in reducing production costs, with an impact of −0.6. Conversely, initial costs emerge as the primary driver increasing production costs, with an impact of 0.6. Other factors, including debt ratio, interest rate, debt term and operation and maintenance (O&M) costs, show marginal impacts ranging from −0.1 to 0.1. Notably, the analysis unexpectedly indicates no apparent impact of electricity export rates on production costs, a finding that warrants further investigation.

3.8. Implications for SMEs adoption

These analyses offer valuable implications for SMEs considering solar PV adoption. The consistently negative NPV underscores the need for additional incentives, cost reductions, or higher electricity rates to enhance project viability. SMEs must carefully evaluate interest rates and electricity prices due to their substantial impact on profitability. The emphasis on reducing initial costs suggests exploring strategies such as bulk purchasing, alternative technologies, or phased implementation. Furthermore, the potential

Table 8. Policy impact analysis.

Challenge/ issues identified	Proposed policy	Expected impact	Alignment to national and global policies/ regulations	Supporting research
1. Negative NPV and lengthy payback periods	Introduce **tax incentives** and **capital subsidies** for SMEs adopting solar PV systems.	Enhanced financial viability for SMEs, accelerating solar PV adoption.	Aligns with **UN SDG 7** (Affordable and Clean Energy) and Ghana's *Renewable Energy Act*, 2011.	(AckahAcep, 2017; G. Government of, 2011; Ministry of Energy, 2021)
2. High initial investment costs	Establish government/ private sector-led **Green Energy Grants** and **subsidized financing schemes/ Low interest loans** for SMEs investing in solar PV systems.	Lowered upfront costs, increasing accessibility for SMEs.	Aligns with **Paris Agreement** goals to promote financial support for renewable energy projects.	(Arthur et al., 2023; Bank of Ghana, 2023; Enbar et al., 2016; Georgitsioti et al., 2019; "Ghana: Electricity Prices," 2025; Omar & Shaari, 2009; Sarzynski, 2009; Sundaram et al., 2024)
3. Sensitivity to interest rates	Negotiate and Implement **government-backed loan guarantees** or interest rate caps for renewable energy investments.	Mitigate financing risks associated with fluctuating interest rates, improving project attractiveness and stimulating increased SME participation in adopting solar PV systems.	Aligns with Ghana's **Renewable Energy Master Plan (2019)** promoting financial mechanisms for clean energy (Energy Commission, 2019).	(Fu & Irfan, 2022; Jiang et al., 2021; Li et al., 2022; Sher & Qiu, 2022; Sun & Sankar, 2022)
4. Minimal impact of electricity export rates on production costs	Revise and implement **feed-in tariffs (FiTs)** that incentivize grid exports from SME solar projects.	Increased revenue streams for SMEs exporting electricity.	Supports best practices in renewable energy financing, as per the **International Renewable Energy Agency (IRENA)** guidelines.	(Brown, 2013; IRENA, 2020; Lu et al., 2019; Qadir et al., 2021; Scarpellini et al., 2021; Taghizadeh-Hesary & Yoshino, 2020; The World Bank & Climate Investment Funds, 2013; Umamaheswaran & Rajiv, 2015)
5. High Operation and Maintenance (O&M) costs and lack of skilled workforce	Establish **subsidized training programs** for solar PV maintenance and create a national certification for solar technicians.	Long-term operational sustainability, improved system reliability, reduced long-term costs and job creation in green sectors.	Aligns with **ILO Green Jobs Initiative** promoting skill development for renewable energy projects	(Couture et al., 2010; Groba et al., 2011; Jenner et al., 2013; Kreycik et al., 2011; Lam & Law, 2018; Scarpellini et al., 2021)
6. Limited economies of scale	Promote **cooperative solar projects** among SMEs to share costs, risks and benefits.	Reduced per-unit costs, improved cost-effectiveness and scalability of solar PV systems, making them more feasible for SMEs.	Reflects best practices in community energy models recommended by **IRENA** and **Ghana Energy Commission.**	(Couture & Gagnon, 2010; Kozar et al., 2022; Lan et al., 2020; Smith & Urpelainen, 2014; Technical et al.)
7. Lack of adaptive financial models	Mandate **periodic reviews of solar financing models** based on market dynamics and SME feedback.	Better alignment of financial models with real-world SME needs, accurate financial planning and increased investor confidence.	Supports Ghana's **Renewable Energy Master Plan** and global recommendations by **IRENA** for adaptive energy policy frameworks.	(Hufen & Koppenjan, 2015; Langthaler et al., 2021; Luke, 2023; Markus, 2019; Repp et al., 2021)
8. Uncertainty in energy export profitability	Encourage **dynamic tariff adjustments** that consider market fluctuations and align with electricity demand cycles.	Improved financial predictability for SMEs, enhancing their confidence in solar investments.	Supports best practices in renewable energy financing, as per the **International Renewable Energy Agency (IRENA)** guidelines.	(Camarinha-Matos et al., 2017; Gennitsaris et al., 2023; Herce et al., 2024; Marques et al., 2023; International Finance Corporation, 2019)
9. Significant GHG emission reductions (93%)	Introduce **carbon credit rewards** or rebates for SMEs achieving emission reduction targets.	Incentivizes emissions reductions, aligning SME practices with environmentally conscious practices.	Aligns with the **Kyoto Protocol** and **Ghana's Climate Change Policy** encouraging carbon mitigation in business sectors.	(Hassan et al., 2023; Topo et al., 2015; Viljoen & Burger, 2014)

(*continued*)

Table 8. Continued.

Challenge/ issues identified	Proposed policy	Expected impact	Alignment to national and global policies/ regulations	Supporting research
10. Dependence on high upfront technology costs	Support **local manufacturing of PV components** to reduce costs through local value chain development.	Reduced reliance on imports, lower PV costs and strengthened local economies.	Supports Ghana's **Renewable Energy Master Plan (2019)** and global recommendations by **IRENA.**	(Del Río, 2012; Dupont et al., 2014; Dutta & Mitra, 2017; Hao et al., 2024)
11. Lack of public awareness on solar PV benefits	Conduct **awareness campaigns** targeting SMEs, emphasizing economic and environmental advantages of solar energy.	Increased SME interest and community support for solar energy projects.	Aligns with the **Kyoto Protocol, Ghana's Climate Change Policy and** with Ghana's **Renewable Energy Master Plan (2019)**	(Anshika, Anukwonke & Abazu, 2022; Blaschke, 2022; Climate Innovation Centre, 2023; IRENA, 2017; Stute & Klobasa, 2024)

for improved viability through economies of scale indicates that SMEs might benefit from larger-scale projects or collaborative initiatives.

The findings also highlight the importance of maximizing electricity export to the grid as a key strategy for reducing production costs. In terms of financing, SMEs might consider leveraging higher debt ratios and negotiating longer debt terms while securing lower interest rates to optimize project economics. Although O&M costs show a relatively small impact, implementing efficient maintenance practices can contribute to overall cost reduction.

The unexpected lack of impact from electricity export rates on production costs in this analysis suggests that SMEs should conduct additional investigations or seek expert consultation to fully understand this aspect's influence on their specific projects. Moreover, the analysis's use of a 25% range for all parameters indicates significant uncertainty, emphasizing the need for SMEs to develop robust contingency plans.

3.9. Policy impact analysis and recommendations

While solar energy emerges as a promising and environmentally beneficial option for SMEs in Cape Coast, Ghana, the economic challenges identified necessitate targeted policy interventions. Policymakers should consider implementing cost-reduction strategies, exploring alternative financing options and providing government subsidies on PV components for SMEs. Additionally, the potential for distributed generation in market-based SMEs should be investigated. Periodic reviews of financial models and incentives are crucial for enhancing the economic attractiveness of solar projects for SMEs. Moreover, continued research efforts to improve PV technology and deployment strategies will contribute to achieving cleaner energy adoption and supporting environmental sustainability goals. This study serves as a vital input for formulating effective energy policies and coping mechanisms to ensure the sustainability of SMEs in Ghana.

Table 8 provides a comprehensive policy framework addressing economic, technical and environmental challenges informed by robust research and global standards to ensure sustainable solar PV adoption for SMEs. These policies aim to provide the government and solar PV system investors with policy requirements that will promote profitability while empowering SMEs to improve the adoption rate of solar PV systems to support Ghana's transition to a cleaner and more resilient energy future.

4. Conclusion

Solar energy is considered the most propitious energy source due to its high reliability and safety. They are also an economic alternative to conventional energy. In this paper, a detailed methodology to design and simulate systems for small businesses in Cape Coast, Ghana, was achieved. The comprehensive case study on empowering small businesses through the maximization of solar sizing potential has provided valuable insights into the feasibility and economic viability of implementing solar power systems for various categories of businesses based on energy requirements. The study aimed to assess the energy needs, economic factors and environmental implications of transitioning to solar power in this region.

The analysis reveals a region with favorable environmental conditions for solar energy generation, marked by consistent temperatures, moderate wind speeds and substantial solar radiation potential, even with seasonal variations. However, the economic assessment, as reflected in the RETScreen financial viability analysis, paints a challenging picture for investors. All three tiers of solar systems exhibit negative Net Present Values (NPV) and annual life cycle savings, suggesting that the initial investment costs outweigh expected future savings. The simple payback periods are lengthy, ranging from 17.8 to 20.3 years, posing high risks and delayed returns. The expansion of the negative zone in the cash flow diagram after 14 years further raises concerns about financial health and resilience, warranting a re-evaluation of the project's financial model.

The sensitivity analysis reveals that while the current project parameters indicate financial challenges, these analyses provide SMEs with crucial insights for optimizing solar PV adoption strategies. By focusing on key drivers such as electricity export maximization and initial cost management, while also considering financing terms and operational efficiencies, SMEs can develop comprehensive approaches to enhance project viability. The results underscore the importance of a holistic strategy that addresses multiple factors to improve the economic feasibility of solar PV projects for small and medium-scale enterprises.

To promote solar photovoltaic (PV) systems in Ghana, several key policies are recommended. Implementing tax breaks and subsidies for small and medium-sized enterprises (SMEs) can lower initial capital costs and enhance financial viability. Establishing feed-in tariffs would provide stable revenue for solar energy producers, encouraging more SMEs to invest. Additionally, developing financing mechanisms like low-interest loans or grants can alleviate high-interest barriers. Capacity building through training programs will enhance technical skills among local businesses, while a streamlined regulatory framework will simplify the permitting process for installations. Initiating public awareness campaigns will educate SMEs about the benefits of solar energy and ensuring alignment with national energy policies will facilitate coordinated efforts in renewable energy adoption. Finally, investing in research and development can improve solar technology efficiency and reduce costs, creating a favorable environment for solar PV adoption that contributes to increased renewable energy capacity and reduced greenhouse gas emissions in Ghana.

Additionally, SMEs could leverage these findings to advocate for more supportive policies, such as improved feed-in tariffs or tax incentives, to further bolster the attractiveness of solar PV projects in the SME sector. The study demonstrated that implementing solar power systems had a significant positive impact on reducing greenhouse gas emissions. Emissions were reduced by 93% across all tiers, which is a substantial environmental benefit. Therefore, while technical performance is encouraging, significant economic barriers need to be addressed through policies that promote cost-reduction strategies, alternative financing options and government subsidies to make such projects financially attractive to investors.

4.1. Further studies

Studies are currently being conducted to evaluate energy consumption in West Africa's largest market, located in the Ashanti Region, which comprises over 8000 stores, as well as the feasibility of adopting solar energy. The Kumasi Metropolitan area, where the market is situated, has a solar irradiation intensity of 1726 kWh/m^2, comparable to that of Cape Coast. Additionally, a parallel study is underway at the Bolgatanga Market in Ghana's Upper East Region, which has one of the country's highest solar irradiation intensities at 2042 kWh/m^2. These investigations aim to assess solar energy's potential to meet the energy demands of these markets.

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Ethical approval

This study received ethical approval from the Ethical Review Committee at Cape Coast Technical University, with approval number CCTUERC/EC/2024/015. Components of the study involving participants were carried out in accordance with the ethical principles outlined in the Declaration of Helsinki.

Consent form

Prior to participation, informed verbal consent was obtained from all participants. Participants were provided with detailed information regarding the purpose of the study, their rights and the confidentiality of their responses. Those who agreed to participate voluntarily provided their consent before data collection commenced.

Author contributions

CRediT: **Kwame Anane-Fenin**: Conceptualization, Methodology, Writing – original draft; **Nana Yaa Serwaah Sarpong**: Data curation, Formal analysis, Writing – original draft; **Theophilus Adu Frimpong**: Formal analysis, Investigation, Writing – original draft; **Mukuna Patrick Mubiayi**: Writing – review & editing; **Richmond Kuleape**: Writing – review & editing; **Damenortey Richard Akwada**: Writing – review & editing; **Wisdom Eyram Kwame Agbesi**: Data curation; **Juliana Sally Renner**: Writing – original draft; **Charles Emmanuel Oppon**: Writing – review & editing; **Lawrence Atepor**: Writing – review & editing; **Felix Menu**: Writing – review & editing.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the Corresponding Author (Kwame Anane-Fenin).

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